

Mirabel Reid

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RESEARCH FOCUS

Postdoctoral researcher in theoretical machine learning, working on mathematically tractable models of neural computation and on the cognitive abilities of large language models, with a broader interest in the interface between machine and human learning. Publications at COLT, AAAI, and CogSci.

EXPERIENCE

Technical University of Darmstadt — Postdoctoral Researcher Mar 2026 – Present
Darmstadt, Germany

- Research on representation learning in the Computational Modelling of Intelligent Systems group, with Prof. Angela Yu.

Meta — Software Engineering Intern May 2025 – Aug 2025
Sunnyvale, CA

- Benchmarked statistical and neural approaches to synthetic tabular data generation.
- Designed a pipeline to train generative models on high-frequency temporal snapshots.

Max Planck Institute for Intelligent Systems — Research Intern May 2023 – Aug 2023
Tübingen, Germany

- Developed models for online human/AI expert deferral under budget constraints; work published as *Online Decision Deferral under Budget Constraints* (see Publications).
- Built a Python simulation framework evaluating deferral policies over real and synthetic human and AI performance data.

Los Alamos National Laboratory — Research Intern May 2022 – Oct 2022
Los Alamos, NM

- Improved metadata management for scientific ML workflows; work published as *Improving Radiography Machine Learning Workflows via Metadata Management* (see Publications).
- Built metadata visualization and selection tooling in Python and SQL with domain scientists.

Software Engineering Institute, CMU — Intern, Emerging Technology Center Jan 2020 – Aug 2020
Pittsburgh, PA

- Investigated Graph Neural Networks for bug detection in software development.
- Prototyped and evaluated models in PyTorch.

EDUCATION

Georgia Institute of Technology, Atlanta, GA Sep 2020 – Dec 2025
PhD in Computer Science. Advised by Prof. Santosh Vempala.
Thesis: *On k-Winners-Take-All as a Model of Neuron Communication*

University of Pittsburgh, Pittsburgh, PA Sep 2016 – Apr 2020
BS, double major in Computer Science and Mathematics.

PUBLICATIONS

Peer-Reviewed Conference Publications

Reid, M., Sharma, K., & Singh, A. (2026). Measuring Algorithm Understanding in Humans and AI. *Proceedings of CogSci 2026*.

Reid, M. & Vempala, S. S. (2025). Does GPT Really Get It? A Hierarchical Scale to Quantify Human vs AI's Understanding of Algorithms. *Proceedings of AAAI 2025*. **Oral presentation**.

Reid, M. & Vempala, S. S. (2023). The k-Cap Process on Geometric Random Graphs. *Conference on Learning Theory (COLT)*, 3469–3509. PMLR. **Oral presentation**.

Preprints and Work Under Review

Reid, M. & Zhang, D. (2025). The α -Cap Process: A Continuous Model for Random Geometric Networks of Binary Neurons. *arXiv:2508.09396*.

Reid, M., Sühr, T., Vernade, C., & Samadi, S. (2024). Online Decision Deferral under Budget Constraints. *arXiv:2409.20489*.

Reid, M., Sweeney, C., & Korobkin, O. (2024). Improving Radiography Machine Learning Workflows via Metadata Management for Training Data Selection. *arXiv:2408.12655*.

TECHNICAL SKILLS

Languages: Python, Julia, SQL
ML & Tools: PyTorch, Git
Areas: Learning theory, random graph models, computational neuroscience, LLM cognition, online decision making

HONORS AND AWARDS

Georgia Tech ARC-ACO Fellowship	Nov 2023
ARC Triad Research Fellowship	Nov 2021
Georgia Tech President's Fellowship	Apr 2020

SELECTED TALKS AND POSTERS

Talks
On k-Winners-Take-All as a Model of Neuron Communication — CMU Theory Lunch Seminar, 2025.
Does GPT Really Get It? — Lightning talk, Simons Institute Workshop on Higher-Level Intelligence, 2024; also presented as a poster at the NeurIPS 2024 Behavioral ML Workshop.

Posters
Modelling Hebbian Plasticity with Synaptic Scaling in a k-Winners-Take-All Network — International Conference on the Mathematics of Neuroscience and AI (neuroMONSTER) 2026.
Assemblies and the k-Cap Process: The Effects of Locality on Neural Firing Dynamics — COSYNE 2023.
Convergence of the k-Cap Process on Graphs with Weight Reciprocity — INFORMS Applied Probability Society Meeting, 2023.

SERVICE AND TEACHING

Reviewer: NeurIPS 2026, ICLR 2025-2026, and ALT 2024-2026.
Teaching Assistant: Advanced Algorithms (Fall 2023); Introduction to Graduate Algorithms (Fall 2021); Discrete Mathematics (Spring 2019).